

Section 14 , in the previous section we trained an ANN model. In this section we will use RNN model

End to End Deep Learning Projects Using Simple RNN

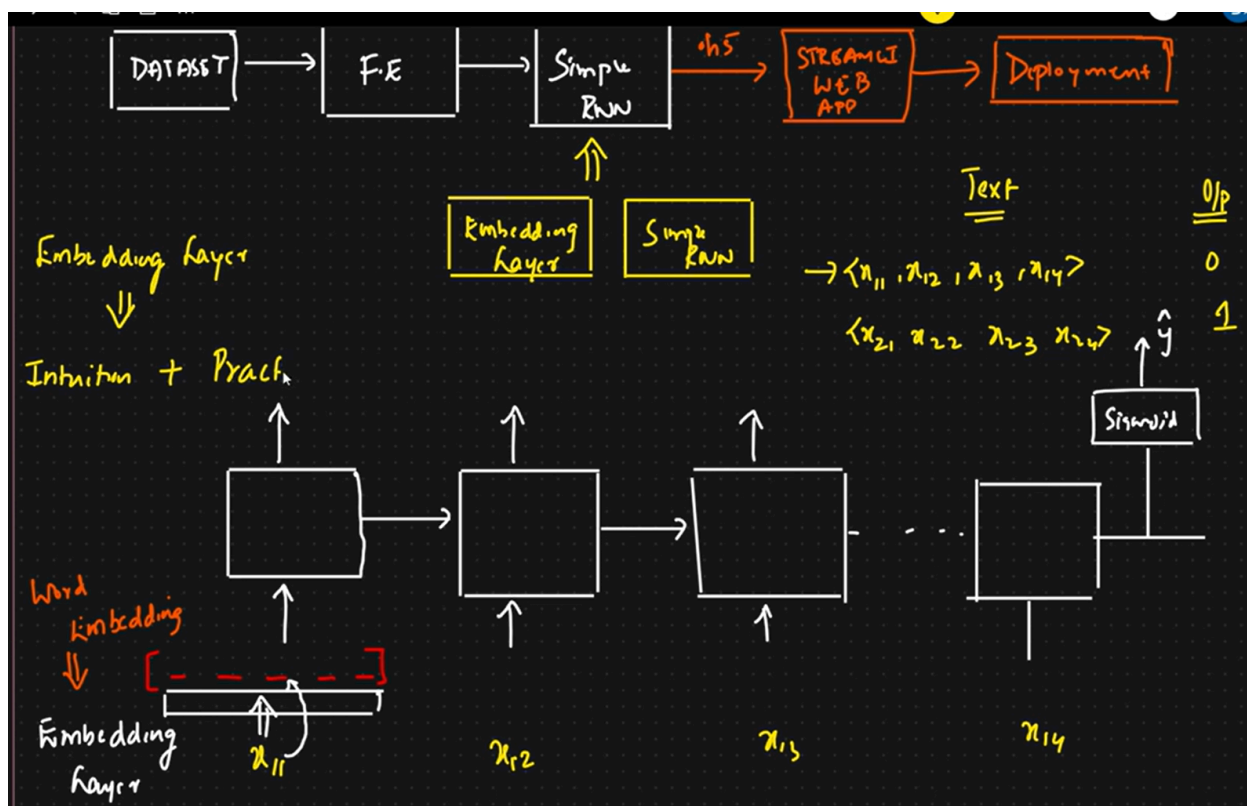
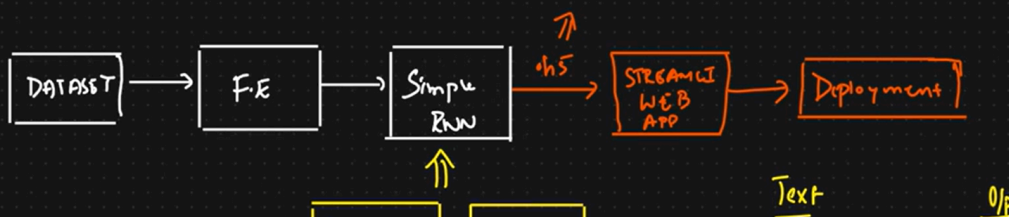
Imdb Dataset [Movie Reviews Dataset] ←

Text
Reviews

O/P

Positive/Negative ⇒ Simple RNN ⇒ prediction ⇒ TRAINING
↓
Deployment

Google Colab ⇒ Check the Model ⇒ Free GPU's

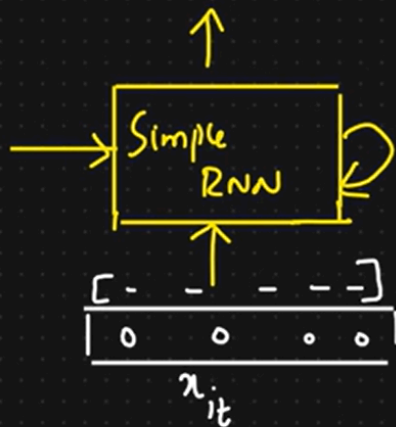


issues with one hot encoding, it is sparse

Word Embedding [Feature Representation] $\Leftarrow$

Dataset

<u>Text</u>	<u>O/P</u>
$\langle x_{11} \ x_{12} \ x_{13} \ x_{14} \rangle$	0
$\langle x_{21} \ x_{22} \ x_{23} \ x_{24} \rangle$	1
- - - - -	-
- - - - -	-



⑧ One hot Encoding

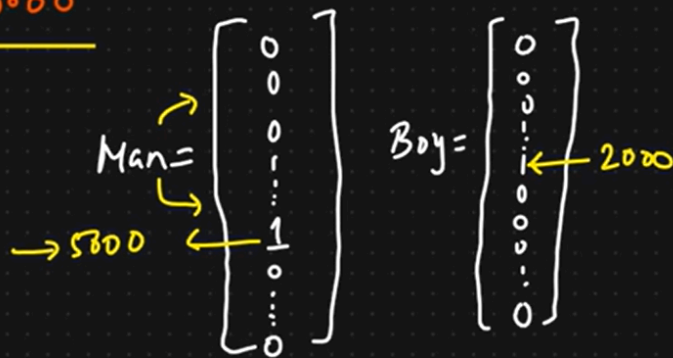
$|V| = 10000$

One hot

⑧ One hot Encoding

$|V| = 10000$

One hot Representation.



$\Downarrow$
Sparse matrix $\Rightarrow$ Overfitting

② Word Embedding		[Word2vec] ← Embedding Layer						$ V = 10000$	
Feature Representation		(2000)	(5000)	(6000)	(7000)	(1000)	(7000)		
		BOY	GIRL	KING	QUEEN	APPLE	MANGO		
300 dimension	Gender	-1	1	-0.92	0.92	0.0	0.1		
	Royal	0.01	0.03	0.95	0.96	-0.02	0.01		
	Age	-	-	-	-	-	-		
	Food	-	-	-	-	-	-		
	...	-	-	-	-	-	-		
300 dimension									

$|V| = 10000$ features - dimension = 300

Practical Implementing : Word Embedding layer ⇒ Some Example

In this tutorial we will understand the embedding layer where the data set(text) is converted to vectors using word2vec, that we discussed in previous video..And also mathematical intuition and practicality. This will be a separate layer

Go to this URL <https://jalammar.github.io/illustrated-transformer/>

- 1.First do token embedding(convert the input text to vector)
2. Linear transformation : create Q,K, V by multiplying the embeddings by learned weights matrices w_q, w_k, w_v
- 3.Compute attention scores: here we multiply the W_q matrices with K matrices got for each word.
- 4.Scaling: We take up the scores from point 3, and scale down by dividing the scores by the root of D_k

<https://www.youtube.com/watch?v=csWluHwfsB8>

<https://www.youtube.com/watch?v=ZhAz268Hdpw>

Learn with Jay

<https://www.youtube.com/watch?v=IRylkiFdUdk&list=PLuhqtP7jdD8CQTxwVsuiFYGvHtFpNhIR>
3